NEWTONITE UI/UX ASSIGNMENT — SPROUT 2.0

Designing Trust & Reducing Friction for First-Time Indian Investors

Author: Product Design & UX Strategy Case Study

Context: Redesigning the web homepage and first 5 onboarding screens under strict 3-week deadlines, device constraints, and stakeholder tensions.

40,000	9,000	3,000
Total Downloads	Completed KYC (22.5%)	First Investments (7.5%)

Executive Summary

Sprout enables first-time retail investors to start their journey from ₹100. However, a severe drop-off occurs between app download and KYC verification (77.5% loss). This case study documents an exhaustive data audit, conflict resolution framework across 5 conflicting stakeholders, WCAG AA accessibility compliance, performance tuning for budget devices under ₹10,000, and a unified mascot-guided onboarding storytelling system.

Sprout 2.0 bridges website discovery and mobile onboarding into a seamless narrative flow. By solving document readiness upfront and replacing ambiguous loading states with transparent status cues, we protect user trust and minimize churn.

1. Data Audit & Flaw Identification

Raw metrics often mask operational realities. Below is our rigorous audit of Sprout's baseline operational data, separating vanity metrics from actionable insights.

Metric	Stated Number	Identified Flaw / Bias	Design Assumption
Homepage Bounce	78% over 2 wks	Skewed by active paid ad campaigns bringing low-intent traffic.	Organic traffic intent is healthier; adjust messaging clarity above fold.
KYC Drop-off	62% drop	Aggregate number lacks step attribution; Support notes missing PAN/Aadhaar readiness.	Users abandon when caught unprepared or when waiting screens look frozen.
Mobile Hardware	91% mobile; 40% <₹10k	Heavy performance and rendering bottlenecks on budget Android devices.	Assets must be ultra-lightweight (<100 KB Lottie/SVG), minimal JS overhead.
Regional Language	"Many" prefer Hindi	Qualitative assertion without exact distribution percentages.	Architecture must support Devanagari font scaling and +30% layout padding.
User Survey	n=40 (65% scared)	Sample restricted solely to existing active users; misses churned drop-offs.	Investment fear is significantly higher among un-onboarded visitors.

The Core Funnel Leak

Out of 40,000 downloads, only 9,000 finish KYC (22.5%), and 3,000 invest (7.5%). The download-to-KYC transition is our primary design target for friction reduction.

2. Problem Statement & User Personas

Problem Statement: *"First-time, mobile-first Indian investors drop off before completing KYC because they feel unprepared, lack necessary document readiness, and misinterpret waiting screens as application crashes —exacerbated by performance bottlenecks on budget devices."*

Persona A: Rohan, 22 — The Budget-Device Fresh Graduate

Context: Living in Tier-2 India, using an Android phone valued around ₹8,000 on a standard 4G connection.

Goals: Wants to start small (₹100) to learn investing safely without risking emergency savings.

Pain Points: Low patience for slow loading times (>6 sec); anxious when verification screens stall; fearful of hidden fee structures.

Persona B: Sunita, 34 — The Regional Language First-Timer

Context: Homemaker / small-business owner in a small town who prefers Hindi or regional terminology.

Goals: Wants transparent, plain-language guidance on how PAN and Aadhaar data are secured.

Pain Points: Intimidated by complex English financial jargon, dense legal disclaimers, and unexpected verification steps.

3. Stakeholder Conflict Matrix & Resolutions

Balancing competing stakeholder demands requires deliberate compromise without diluting core UX principles or violating regulatory constraints.

Stakeholder	Demand / Input	Underlying Concern	UX Resolution & Compromise
CEO	Apple-like minimal, premium aesthetic; zero cartoon characters.	Brand credibility and trust for first-time investors.	Adopted a geometric, abstract sprout/seedling guide rather than an animated cartoon character.
Head of Marketing	Mascot, video, live ticker, 3 pop-ups, and referral banner on homepage.	High engagement and viral acquisition channels.	Retained the geometric mascot as an onboarding guide; moved marketing pop-ups to clean modular cards below fold.
Compliance Officer	Mandatory 12px risk disclaimer visible above fold; ban "guaranteed", "safe", "risk-free".	Regulatory adherence and legal risk mitigation.	Placed a persistent 12px notice at the top of viewports using approved neutral phrasing.
Lead Developer	Page weight <1.5 MB; animations <100 KB; 3-week timeline, 1 frontend dev.	Technical feasibility and performance on budget phones.	Utilized lightweight CSS transitions, single SVG seedling graphics, and zero heavy video assets.
Support Lead	Clear upfront messaging about PAN/Aadhaar and explicit waiting state indicators.	Reducing support ticket volume and panic drop-offs.	Added a dedicated "A little preparation" screen outlining document needs beforehand.

4. Mascot & Visual Direction

To satisfy the CEO's demand for premium professionalism and Marketing's need for emotional engagement, we designed **The Sprout Guide** — a geometric, abstract dual-leaf seedling resting on a soft organic base.

The 4 Core Mascot States

- **Welcome:** Standing upright, greeting the user with open-ended guidance and language selection.
- **Explaining:** Pointing toward document preparation requirements with checkmark indicators.
- **Waiting / Loading:** Gentle pulsing animation (<100 KB SVG/Lottie) assuring backend verification is active.
- **Celebrating:** Expanding slightly upon successful first investment or KYC milestone.

Design System Foundations

- **Color Palette:** Forest Green (`#11382A`) for high contrast text/buttons; Soft Sage (`#F4F6F0`) for calm background tone; Crisp White (`#FFFFFF`) card surfaces.
- **Typography & Scaling:** Mukta / Noto Sans font stack with support for Hindi Devanagari text expansion (+30% layout padding allowance).
- **Accessibility:** All color combinations meet WCAG AA standards (minimum 4.5:1 contrast). Tap targets strictly $\ge 44\text{ px}$.

5. Homepage Redesign & Storytelling Journey

The homepage transitions the user smoothly from discovery to onboarding while strictly adhering to compliance mandates and performance limits.

Key Homepage Architecture Elements

- **Persistent Risk Disclaimer:** Positioned immediately above the fold in 12px text: *"Investments are subject to market risks. Read all related documents carefully."*
- **Value Proposition:** *"Your investing story starts with ₹100. A starting amount. Not a commitment."*
- **Step-by-Step Clarity:** Three distinct phases outlined clearly: *01 Prepare*, *02 Connect*, *03 Choose your next step*.
- **FAQ Section:** Addresses upfront questions regarding KYC security and market risk without burying information.

Both mobile and desktop layouts maintain visual consistency, utilizing generous whitespace and zero heavy banners to ensure load times remain under 6.2 seconds on 4G networks.

6. App Onboarding: Screens 1 to 3

The first three onboarding screens establish immediate trust, prepare the user for verification, and prevent drop-offs caused by unexpected friction.

Screen 01: Welcome & Language (SETUP 01 / 05)

Introduces the Sprout guide. Features a clean language preference selector (English active, Hindi coming soon) and explicit assurance: *"You're setting up your account. No investment is made."*

Screen 02: Document Preparation (SETUP 02 / 05)

Directly resolves Support complaint #1. Outlines what to keep nearby before starting (PAN card, Aadhaar, Aadhaar-linked mobile) with a reassuring exit option: *"Not ready yet? You can come back later."*

Screen 03: Mobile & OTP Verification (SETUP 03 / 05)

Confirms mobile number linked to Aadhaar. Features 6-digit OTP input boxes with clear validation states and countdown timers for code resends.

7. App Onboarding: KYC & Waiting States

The final onboarding steps handle sensitive identity verification with utmost transparency to prevent panic and app abandonment.

Screen 04: KYC Details (SETUP 04 / 05)

Collects 10-character PAN and 12-digit Aadhaar. Includes a prominent *"Why we ask"* collapsible drawer explaining that verification does not execute an immediate investment.

Screen 05: Verification & Waiting State

Directly resolves Support complaint #2. Replaces blank frozen screens with an active status tracker and explicit reassurance: *"You're setting up your account. No investment is made."* This eliminates crash anxiety during backend API checks.

Design Impact on Funnel

By frontloading document preparation and clarifying loading states, projected KYC completion rate increases from 22.5% toward 45%.

8. Compliance & Deliverables Checklist

All design assets and code specifications have been rigorously audited against project constraints and regulatory guidelines.

- ✅ **WCAG AA Contrast:** All text and UI component color pairings exceed 4.5:1 contrast ratio.
- ✅ **Tap Targets:** Minimum 44px height and width enforced across all buttons and inputs.
- ✅ **Disclaimer Placement:** 12px risk disclaimer visible above the fold on all investment views.
- ✅ **Banned Words Banned:** Zero usage of "guaranteed", "safe", or "risk-free" across copy.
- ✅ **Hindi Text Expansion:** Layouts engineered with +30% horizontal padding for Devanagari localization.

Technical Performance Budgets

- **Page Weight:** Maintained under 1.5 MB for fast 4G loading.
- **Animations:** Pure CSS or Lottie/SVG under 100 KB.
- **Timeline:** Fully deliverable by 1 frontend developer within 3 weeks.

9. Success Metrics & What to Test Next

To evaluate the efficacy of the Sprout 2.0 redesign post-launch, we have defined key quantitative success metrics and iterative testing milestones.

Primary Success Metrics

- **KYC Conversion Lift:** Increase download-to-KYC completion from 22.5% to >40% by reducing document drop-off.
- **Time-to-KYC Reduction:** Decrease average onboarding duration from 6 minutes to under 3.5 minutes.
- **Support Ticket Reduction:** Lower PAN/Aadhaar readiness and crash-panic tickets by 50%.

What to Test Next (Iteration Roadmap)

- **Full Hindi Rollout:** A/B test native Devanagari UI copy against English in Tier-3 target regions.
- **DigiLocker Integration:** Test one-tap document fetching to completely eliminate manual PAN/Aadhaar typing.
- **Interactive Micro-Tutorials:** Test a 30-second animated primer for absolute beginners who report "investing feels scary".